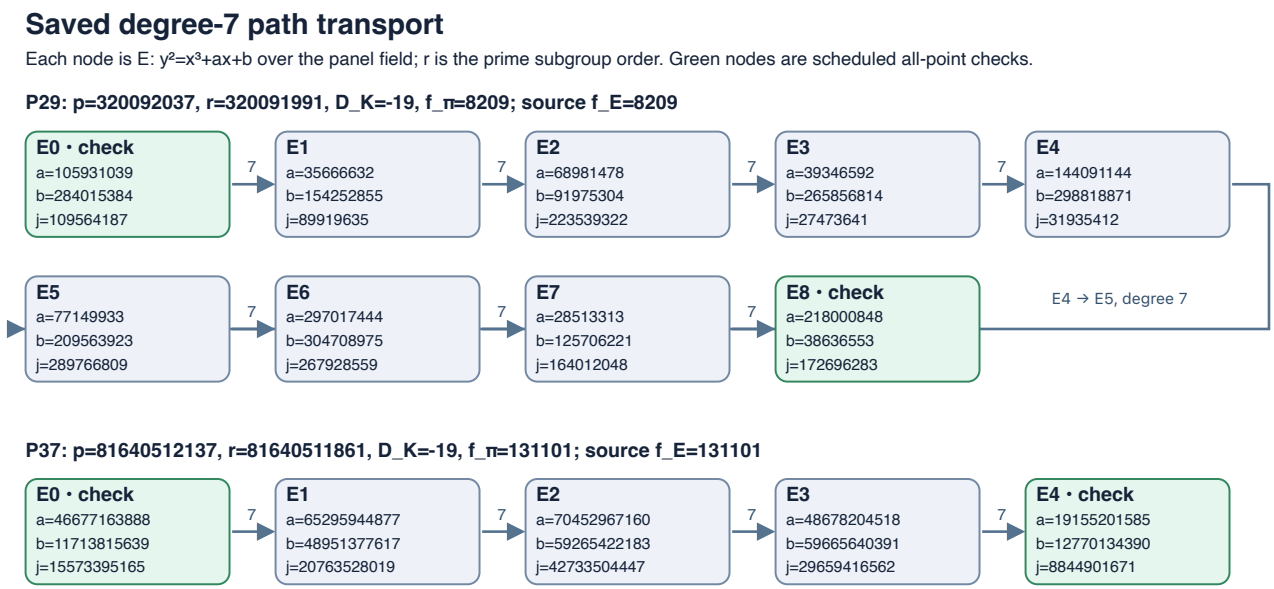
First check-vertex path transport

16 October 2026 — implementation control, T-3361 (successor to T-3350)

The archived P29 and P37 degree-7 walks carried the fixed source `low_x` signed base to their first scheduled all-point check vertices. This report records implementation observations; it assigns no scientific verdict.



Stratum	Edges	Point images replayed	Final checkpoint SHA-256
P29	8	4,096	`8b40a5c11de49dbf…`
P37	4	12,800	`4b529e7d45b6cc54…`

The table shows SHA-256 prefixes for legibility. Complete hashes appear in `insight-20261016.md` and the verifier stdout files.

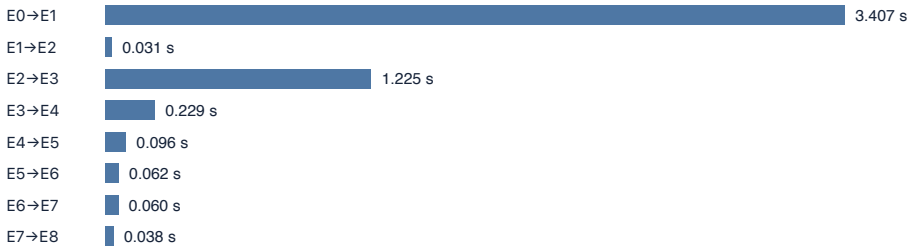
The integer-arithmetic verifier exited 0 on both chains. It replayed all stored base and generator images, the raw codomain identities, the source base definition, the hash chain, and the fixed check-vertex schedule. P29 index 8 and P37 index 4 each retained the full number of distinct `x` orbits. Evidence: `controls/p29-check1-verify.stdout`, `controls/p37-check1-verify.stdout`, and the saved `edge-*.json` files.

Diagnostic execution times

Per-edge control elapsed time

Seconds inside transport_path.py. Diagnostics from one run per edge; not a matched performance comparison.

P29 (n=8 edges; axis 0–4 s)



P37 (n=4 edges; axis 0–20 s)



No CPU, RSS, hardware isolation or uncertainty estimate was recorded. The bars are not an ECDLP cost result.

The times are single internal monotonic measurements made while preparing each edge. Processor isolation, CPU time, RSS, complete shell wall time, and uncertainty were not captured. These bars make no performance or ECDLP cost comparison.

Failure and next gate

The first P29 continuation exited 1 because the Sage launcher redirected its cache to a protected directory. Its stderr is retained as `controls/p29-resume-to-check1.stderr`. The corrected Sage-shell invocation for P29 and the P37 continuation exited 0. Full-path transport, relation collection, rank, and same-point solving remain next-stage work.

The dated prose report `insight-20261016.md` gives exact commands, input hashes, source revision and dirty state, the independent checks, and limits. The editable diagram source is `render_report.py`.